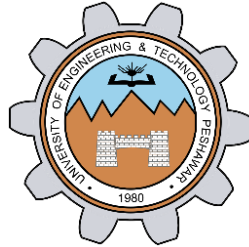


Interfacing ADC with 8051 Microcontroller

LAB # 11



Spring 2024

CSE-307L Microprocessor Based System Design Lab

Submitted by: **Ali Asghar**

Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

23rd May 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Task 1:

From temperature sensor (LM35) read temperature and convert it into Digital value by using ADC0804 and display the value on the LCD. In LCD at first line write your registration Number and on the second line display the value of the temperature sensor attached with ADC0804.

Code:

```
lab_task.c  STARTUP.A51
1  #include<reg51.h>
2  //RS = 1 we are passing data
3
4  sbit RS = P3^0;
5  sbit Enable = P3^1;
6
7  sbit READ = P3^3;
8  sbit WRITE = P3^4;
9  sbit INTR_FLAG = P3^2;
10
11 void init();
12 void delay(unsigned int);
13 void wr_cmd(unsigned char);
14 void wr_data(unsigned char);
15 void display_temp();
16 int ADC();
17 int reading;
18 int temp;
19
20 int hund, ten, unit;
21
22 char *reg_no = "21PWCSE2059";
23
24 void main(void){
25     P0 = 0x00;
```

```
lab_task.c  STARTUP.A51
25     P0 = 0x00;
26     P2 = 0x00;
27     init();
28
29     while(*reg_no)
30         wr_data(*reg_no++);
31
32     while(1){
33         reading = ADC();
34         // reading = reading<<1;
35
36         display_temp();
37     }
38 }
39
40
41 void init(){
42     wr_cmd(0x38); //2 lines and 5x7 Matrix
43     wr_cmd(0x0e); //Display on and cursor blinking
44     wr_cmd(0x01); //Clear Display Screen
45     wr_cmd(0x06); //Increment Cursor
46     wr_cmd(0x0c); //Display on, Cursor off
47     wr_cmd(0x80); //Force cursor to begining of 1st line
48
49 }
```

```
50
51 void wr_cmd(unsigned char cmd){
52     P2 = cmd;
53     RS = 0;
54     Enable = 1;
55     delay(10000);
56     Enable = 0;
57 }
58
59 void wr_data(unsigned char dt){
60     P2 = dt;
61     RS = 1;
62     Enable = 1;
63     delay(10000);
64     Enable = 0;
65
66 }
67
68 void delay(unsigned int n){
69     unsigned int i;
70     for(i = 0; i<n; i++);
71 }
72
73 int ADC(){
74     -----
75 }
```

```
lab_task.c  STARTUP.A51

73 int ADC() {
74     READ = 1;
75     WRITE = 0;
76     delay(10);
77     WRITE = 1;
78
79     while(INTR_FLAG == 1);
80
81     READ = 0;
82     temp = P1;
83     delay(10);
84     return temp;
85
86 }

88 void display_temp() {
89     wr_cmd(0xC0); //Move to 2nd line
90
91     hund = reading/100;
92     reading = reading%100;
93
94     if(hund != 0) {
95         wr_data(hund + '0');
96     }
97
98     ten = reading/10;
99     reading = reading%10;
100
101     wr_data(ten + '0');
102
103     unit = reading;
104     wr_data(unit + '0');
105
106 }
107
```

Proteus Simulation:

